Let's verify your QLSC 612 software setup!

QLSC 612 Spring 2026

Having trouble?

If your local setup doesn't match the expected output, revisit the course setup instructions for your OS:

https://neurodatascience.github.io/QLS612-Overview/setup/setup.html

(Windows users) WSL version and distro

1. Open your **command prompt** and type: `wsl -l -v`
    - In the list of installed distros, you should see:
      ```
      NAME              STATE        Version
      * Ubuntu          Running      2
      ```
      (Your Ubuntu might have a version number)

2. Open your **WSL2 Ubuntu terminal** and type: `lsb_release -a`
    - The Ubuntu you are running should be version `22.04 LTS` or higher

Troubleshooting

- Have multiple Ubuntu versions and running the wrong one? Change the default distro by running in your **Command Prompt:** `wsl --set-default Ubuntu-<VERSION>`
    - Close and relaunch your **WSL Ubuntu terminal** for this to take effect

(Windows users) WSL version and distro

Troubleshooting

- Forgot the password for your WSL2 Ubuntu user account?
    1. Open **Command Prompt** and type:
       ```
       wsl -u root
       passwd USERNAME
       ```
        - If you have multiple Ubuntu distros installed, you will need to specify the distro:
          ```
          wsl -d <distro-name> -u root
          ```
    2. Type `exit`
    3. Close and relaunch your WSL2 Ubuntu terminal

(Windows users) WSL version and distro

Make sure you are using the **WSL2 Ubuntu terminal**, ***not*** PowerShell or Command Prompt, for all remaining check steps (and throughout the course!)

- PowerShell is *not the same* as the WSL2 Ubuntu terminal and won't have access to the same software/files/folders

Bash shell

1. Open a **terminal** and type: `echo $0`
   Expected output: `/bin/bash`

Troubleshooting

- Mac/Linux: You may have to type `bash` first to access the bash shell
  - **Recommended:** change your default shell by typing `chsh -s /bin/bash` (close and reopen your terminal for changes to take effect)

- Windows: Ensure you are in the WSL Ubuntu terminal

Git

1. Type: `git --version`
   Expected output: `git version 2.xx.x`

2. Check your git configuration: `git config --list`
   Look for the following lines in the output:
   ```
   user.email=YOUREMAIL.com
   user.name=YOUR NAME
   core.autocrlf=true
   init.defaultbranch=main
   pull.rebase=false
   ```

3. Check that you can connect to GitHub using the SSH key you set up:
   ```
   ssh -T git@github.com
   ```
   After entering the passphrase for your SSH key, you should see:
   ```
   Hi USERNAME! You've successfully authenticated, but GitHub does not provide shell access.
   ```

Python

1. Type: `conda env list`
   - You should see `qlsc612` in the list of available conda environments.
2. Activate the `qlsc612` conda environment: `conda activate qlsc612`
   - You should now see `(qlsc612)` at the beginning of your shell prompt.
3. Check the path of the Python interpreter being used: `which python`
   - Linux/WSL: `/home/USERNAME/miniconda3/envs/qlsc612/bin/python`
   - Mac: `Users/USERNAME/miniconda3/envs/qlsc612/bin/python`

Tips

- To deactivate a conda env after you're done with it: `conda deactivate`

Docker

1. If you're on Mac or Windows, start the Docker Desktop application.
2. In a **terminal**, type: `docker run hello-world`
   You should see a message that starts with:
   ```
   Hello from Docker!
   This message shows that your installation appears to be working correctly.
   ...
   ```

Troubleshooting

- Mac/WSL: If you get a "command not found" error when running Docker commands, always check that Docker Desktop is running

Getting the course materials from GitHub

For convenience, we will store the course materials in our home directory (`~`).

1. In a **terminal**, run these commands:
   ```
   cd ~
   git clone git@github.com:neurodatascience/QLS-course-materials.git --depth 1
   ```
   This will take a few seconds.

2. Once done, run: `ls`
   - You should see a folder `QLS-course-materials` among the output

Tips

- If you've already cloned the repo elsewhere, we recommend deleting it and following the above steps to get the latest version of the materials in your home directory

Updating course materials from GitHub

From now on, to ensure you have the latest version of lecture materials from GitHub, simply type:

```
cd ~/QLS-course-materials
git pull
```

Note for Windows users

- `QLS-course-materials` will not be accessible in your normal file explorer 📁 because the WSL filesystem is **separate** from your normal Windows filesystem
- In this course, we will only be working with `QLS-course-materials` through the WSL2 Ubuntu

VS Code + Jupyter notebook plotting

1. Ensure that the `qlsc612` environment is active: `conda activate qlsc612`
2. Open your current directory in VS Code: `code .` (period)
3. In VS Code, open a terminal by pressing `` CTRL+` `` (backtick)
4. In the **VS Code terminal**, open a new Jupyter Notebook file: `code plot_test.ipynb`
5. In the top right corner, click `Select Kernel`, then choose from the Python Environments `qlsc612 (Python 3.12.x)`

VS Code + Jupyter notebook plotting

6. Press `+Code` at the top of notebook to add a code cell
7. Type this code in the cell:

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots()
ax.plot([1, 2, 3], [1, 4, 2])
plt.show()
```

8. Press ▶ next to the cell to run the code (or `CTRL+Enter` / `CMD+Enter`).